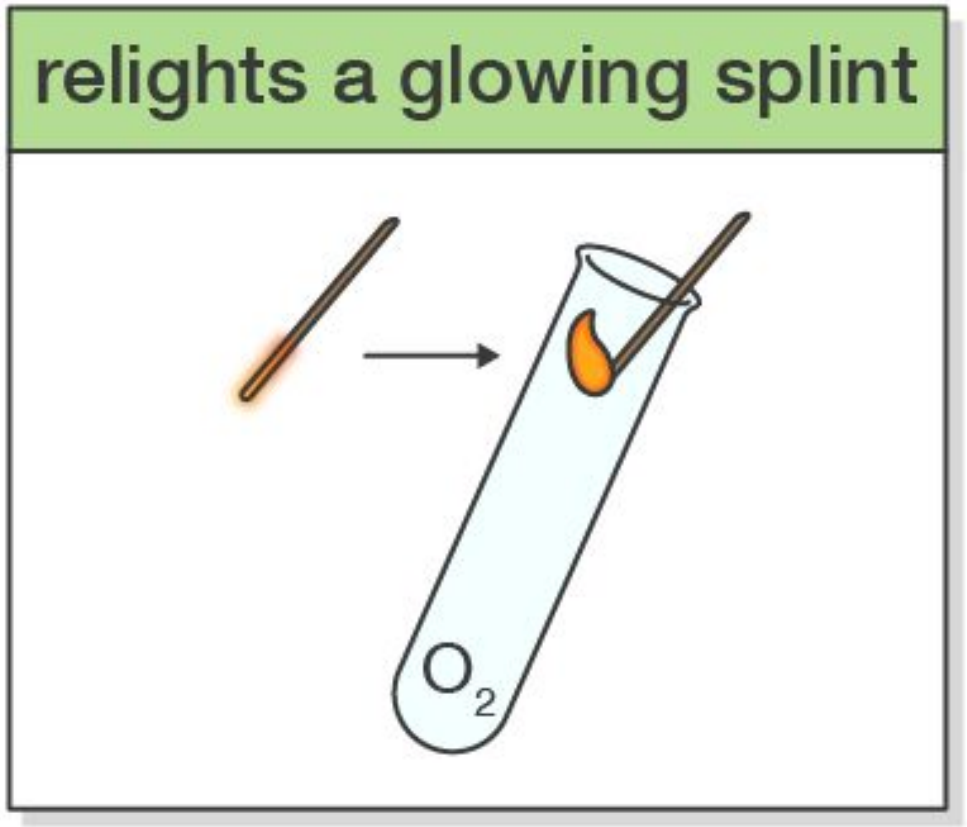
The Gas Tests

Properties of Gases

- Most gases are **colourless** and **odourless** – so how can we identify them?
- Although the may have similar **physical** properties, their **chemical** properties differ
- These properties can be used to **identify** some common gases

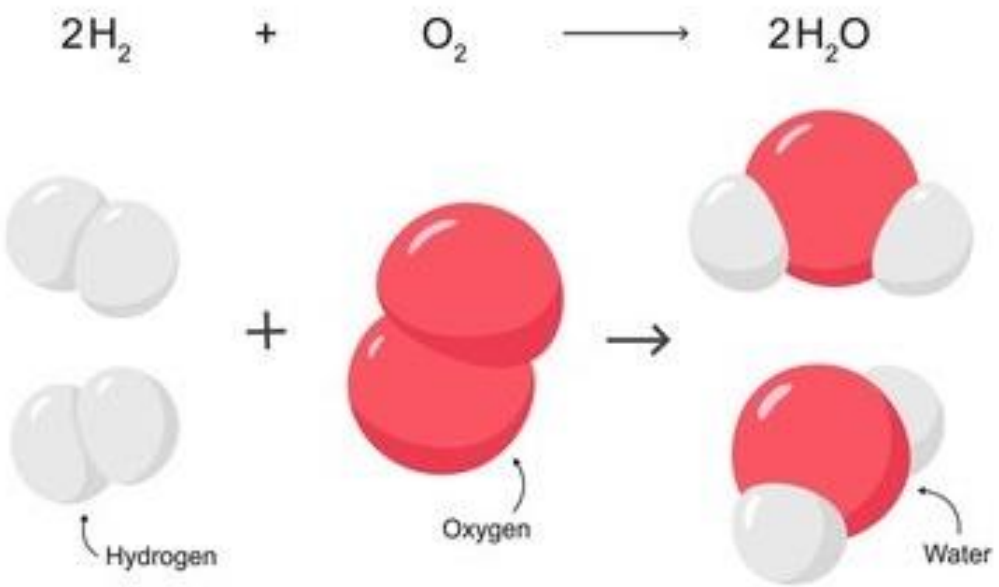
Oxygen

- Oxygen is essential for **combustion**
- A **glowing splint** test is used to identify it
- If the glowing splint **relights** , then oxygen is present



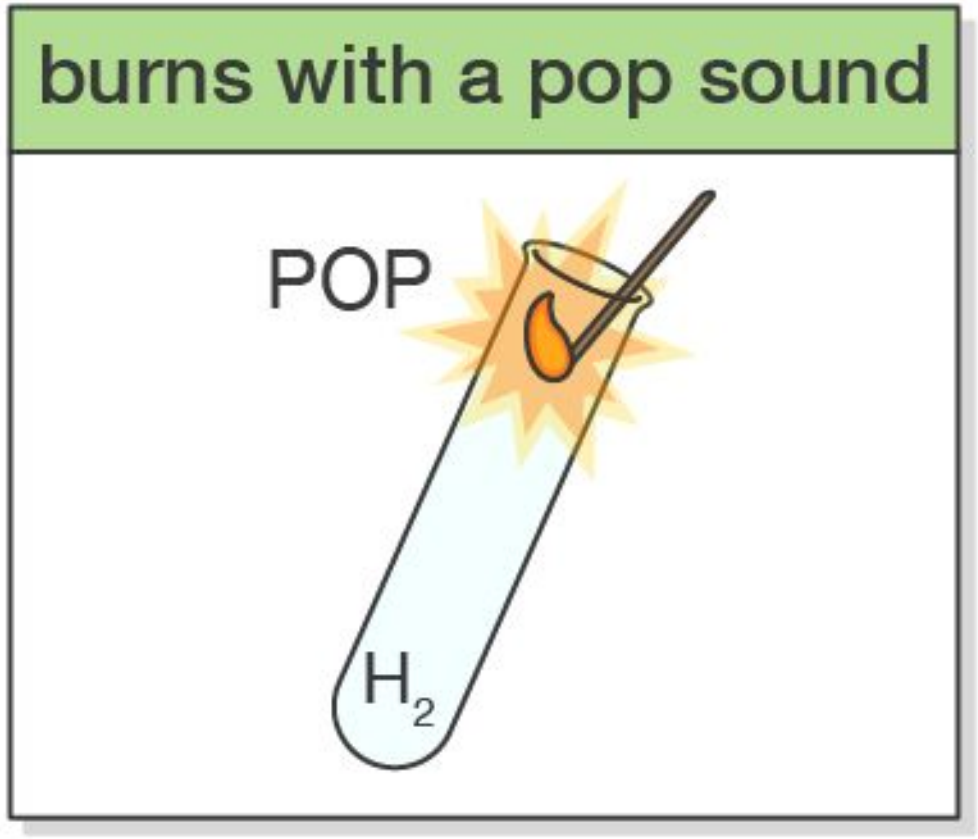
Hydrogen

- Hydrogen is highly **combustible**
- When hydrogen is **ignited** , the hydrogen atoms combine with oxygen in the air to form **water** molecules



Hydrogen

- This rapid reaction is often accompanied by an explosive "**pop**" sound
- A **flaming splint** test is used to identify it



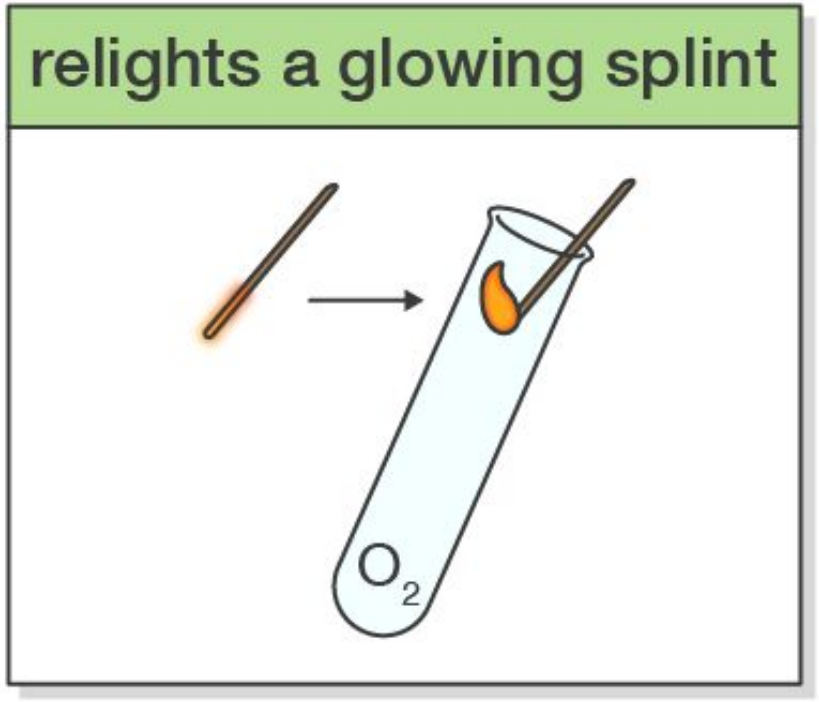
Carbon Dioxide

- Carbon dioxide does **not** support combustion
- It will put **out** a flame (extinguish), which is why it is found in most types of **fire** extinguishers

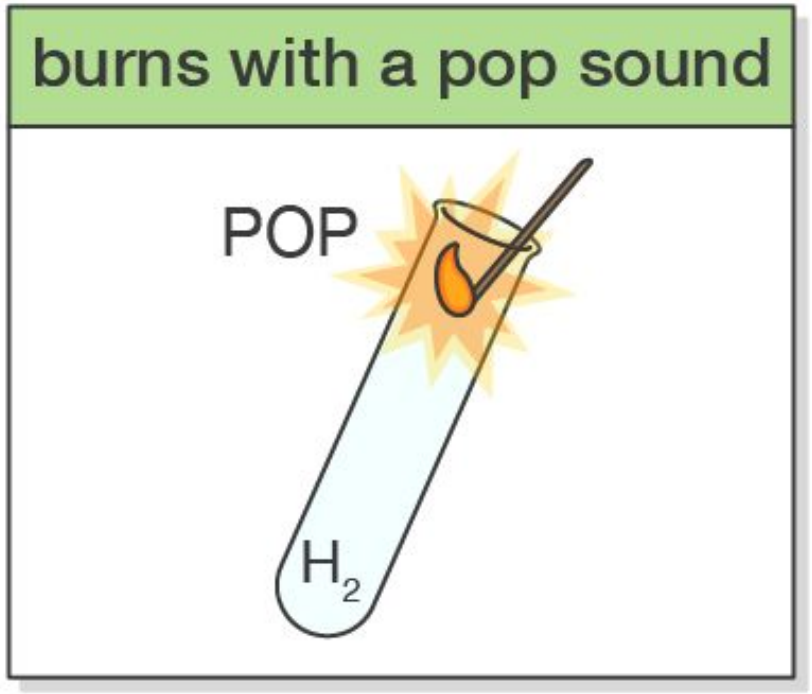
Gas Test Lab

- Follow the instructions CAREFULLY
- Remember the Rules of Lab Safety
- Record your observations accurately
  - Don't write what you THINK should happen, write what DID happen

Oxygen



Hydrogen



Carbon dioxide